

Input and detection

Frames I_{k-1}, I_k



MOT17-13 · moving camera



MOT20-02 · dense crowd



UAVDT M0101 · aerial



KITTI 0011 · translating camera + GPS/IMU



Detector, run once

YOLOX-X (MOT17)
COCO-pretrained (KITTI)



Detections D_k

Byte-identical across all configurations; SHA-256 per file

Instrumented online compensation

output W_k



Shi-Tomasi



pyramidal LK

RANSAC



4-DOF similarity

Recorded: inlier mask, residuals, counts, positions; returned warp unchanged (max $|\Delta H| = 0$)

ρ

n_{in}

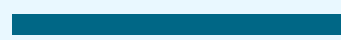
ε

π

τ

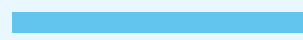
φ

ε alone



0.866

all six



0.774

held-out AUC, leave-one-sequence-out



Oracle-warp contrast and gate flips

output W_k^*

Full-resolution SIFT

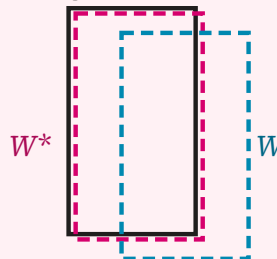
Mutual NN + ratio test

Annotated foreground masked

Forward-backward check

Non-causal reference

ground truth



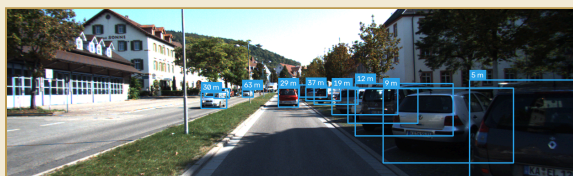
Warp the previous ground-truth box by both warps; count pairs whose gate decision differs:

$$(1 - \text{IoU}_{\text{on}} > \theta) \oplus (1 - \text{IoU}_{\text{ref}} > \theta)$$

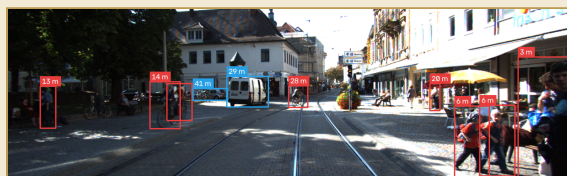
55 of 109,955 pairs flip; 34 harmful

Depth-aware global homography

output H_k



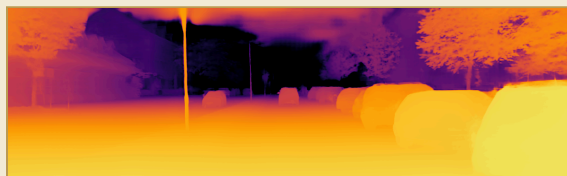
0011/131 · 5–63 m · 114.1 → 3.27 px



0013/88 · 3–41 m · 116.6 → 8.94 px



Deployed GMC rejects near points (red)



far near
Depth-Anything-V2 Metric $\hat{Z}$, 4.1–79.6 m

- 1 Back-project $X = \hat{Z} K^{-1} \tilde{x}$
- 2 Ego-motion (R, t) from GPS/IMU
- 3 Re-project $x' = \pi(K(RX + t))$
- 4 Fit one global H_k

Within-frame spread, KITTI (px)

deployed GMC 8.67

fitted similarity 2.43

depth-aware H 1.37

Controls: per-target correction shuffled placebo depth-noise sweep σ

Fixed tracker and evaluation

BoT-SORT association; only the warp is substituted

Kalman prediction $\hat{x}_{k|k-1}$



Warp prediction

W_k

W_k^*

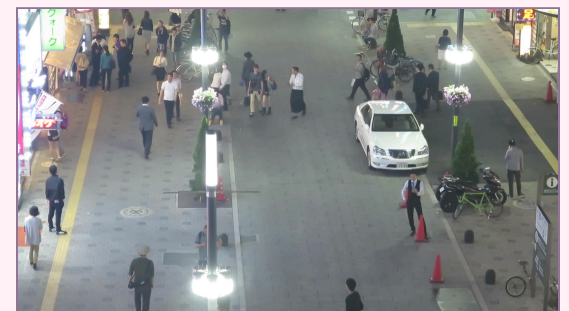
H_k



IoU gate $\theta_{\text{IoU}} + \text{ReID cost}$



Hungarian assignment



MOT17-04, static camera: control for the oracle-warp instrument



TrackEval

HOTA · IDSW · bootstrap 95 % CI over sequences

Input and detection

Frames from MOT17, MOT20, UAVDT and KITTI pass through the detector once. Every configuration reads the same stored detections, verified by SHA-256.

Online compensation

BoT-SORT's sparse-flow GMC, reimplemented to record its inlier statistics. Six reliability signals are compared; the median transfer residual ε alone gives the highest held-out AUC.

Oracle contrast

A stronger non-causal reference warp W^* measures online error and is substituted into the tracker. Gate-flip counting records which association decisions that error changes.

Depth-aware H

On KITTI, background points at monocular depth are moved by sensor ego-motion and fitted by one global homography H . Per-target, placebo and depth-noise arms serve as controls.

Tracker and evaluation

A fixed BoT-SORT pipeline receives one of the three warps. Outputs are scored with TrackEval (HOTA, IDSW) and bootstrap intervals over sequences.